Munasib Khareed - An AI Driven E-Commerce Decision Support System

Final Year Project Proposal

Session FALL 2026

Bachelors of Science, Computer Science



Department of Computer Science

Fast School of Computing
FAST National University, Karachi Campus

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Project Registration

Project ID (for office use)	
Type of project	[✓] Traditional [ ] Industrial [ ] Continuing
Nature of project	[ ] Development [✓] Research and Development [ ] Research
Sustainable Development Goals (SDG's)	[ ] Good Health and Well-Being [ ] Quality Education [✓] Industry, Innovation and Infrastructure [ ] Gender Equality [ ] Decent Work and Economic Growth [ ] Climate Action
Area of specialization	[✓] Artificial Intelligence(AI) [✓] Data Science and Analytics [ ] Internet of Things(IoT) [ ] Blockchain [ ] Mobile App Development [✓] Web Development [ ] Cyber Security [ ] Game Development [✓] Natural Language Processing (NLP) [ ] Other

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Declaration: FYP group members have cleared all prerequisite courses for FYP-I as per their degree requirements

1. Project Abstract

Munasib Khareed is an AI-driven e-commerce decision-support system designed to help consumers make purchase decisions by analyzing available product and customer-review evidence. Unlike recommendation systems that primarily suggest products based on popularity, similarity or ranking the proposed system acts as a virtual consumer advocate. Its purpose is to investigate a selected product identify concerns evaluate the reliability of available evidence and provide an explainable consumer verdict.

The system will utilize a category from the existing UCSD Amazon Reviews dataset. Relevant information such as product metadata, category, brand, description, price, sales rank, customer reviews, ratings, helpful votes, review timestamps, verified-purchase information and related products will be analyzed. The system will consist of Product Intelligence, Review Intelligence, Consumer Risk Assessment, Decision Orchestration, Explainable AI, Alternative Product Intelligence and Consumer Advocate Dashboard modules.

Natural Language Processing and machine-learning techniques will be used to analyze customer reviews, including sentiment and positive/negative product aspects. Fuzzy Logic will be incorporated to handle uncertainty, in consumer evidence while the TOPSIS multi-criteria decision-making technique will aggregate evaluation criteria. SHAP will provide explanations showing which factors contribute to the generated consumer-risk assessment.

The expected outcome is a decision-support system that produces a consumer risk score, evidence-strength assessment, major concerns, relevant alternatives and an explainable final consumer verdict. The project aims to demonstrate how AI, NLP, reasoning, multi-criteria decision analysis and explainable AI can be combined to transform raw e-commerce information into understandable and evidence-based purchase guidance.

2. Introduction

a. Research Domain

The project is about Artificial Intelligence. How it works with Natural Language Processing and Machine Learning. It also has to do with AI and E-Commerce Decision Support Systems. Online shopping sites give people a lot of product details. These details include ratings, reviews, prices, product descriptions, rankings and other similar products. Even though this information is helpful people often have to look at different sources before they decide if a product is right for them.

Traditional systems that recommend products usually try to find items that a person might like.. Just suggesting a product does not always explain its weaknesses, how reliable the customer reviews are or the things that might make buying it risky.

b. Importance of the Domain

Customer reviews and product details hold information about what real customers have experienced.. A lot of reviews make it hard to look at them all. Reviews can have both bad comments and the overall score of a product might not show the full picture or the latest opinions.

An intelligent decision-support system can look through all this information. Share the results in a clear way. Of only suggesting a product this system can help people know what they need to know before they make a purchase.

c. Current Situation

Most online shopping systems offer product rankings, ratings, suggestions, summaries of reviews and related products. These tools may still need people to understand the information on their own.

The system we are creating will bring together different pieces of information into a way that is easy for customers to understand. It will look at product quality the feelings in reviews how ratings are spread out how helpful the reviews are, if the purchases are verified, how the reviews change over time prices and similar products.

d. Relevance of AI and Data-Driven Research

There is an amount of text and numbers in online shopping data. This makes the problem perfect for computer analysis. Natural Language Processing can find information from customer reviews. Machine learning can find the feelings and topics in the reviews. Decision-making methods can look at many different factors.
Also Explainable AI can make the final result easier to understand by showing the reasons behind the consumer-risk score. This is important because a decision-support system, for customers should not just give an answer but also explain why it is the answer.

e. Current Work and Proposed Contribution

i. Current Work in E-Commerce Decision Support

The Comparison can be found in "TABLE 1"

TABLE 1. COMPARISON OF CURRENT AND PROPOSED CONTRIBUTION

System	Product Discovery / Recommendation	Review Analysis	Price Analysis	Consumer Risk / Evidence Assessment	Explainability / Guidance	Alternative Products
Amazon (Alexa for Shopping)	Yes; conversational product research and recommendations	Yes; AI Review Highlights and review/aspect insights	Yes; price history and price alerts	No dedicated consumer-risk/ evidence-strength score described in the cited features	Yes; conversational answers, comparisons and shopping guidance	Yes; recommendations and alternatives
Google Shopping	Yes; product discovery, relevance ranking and recommendations	Yes; ratings, reviews, review filters and AI-supported shopping guidance	Yes; price tracking and price insights where available	No unified consumer-risk score described in the cited features	Yes; AI-generated shopping insights and comparisons	Yes; broad product discovery across sellers
Keepa	Limited; primarily focused on tracking and deal discovery	No dedicated review-risk analysis described	Yes; extensive Amazon price history and price-drop alerts	Price-focused rather than consumer-evidence risk assessment	Yes; price-history visualization	Limited; primarily price/deal tracking

System	Product Discovery / Recommendation	Review Analysis	Price Analysis	Consumer Risk / Evidence Assessment	Explainability / Guidance	Alternative Products
Munasib Khareed (Proposed)	Not primarily a recommendation engine; alternatives are provided for decision support	Yes; sentiment, aspects, rating distribution, helpfulness, verification and temporal patterns	Price position is analyzed; future price prediction remains outside scope unless the dataset supports it	Yes; Consumer Risk Score and Evidence Strength using multiple indicators	Yes; Fuzzy Logic, TOPSIS and SHAP-based explanations	Yes; related alternatives are evaluated using the same evidence framework

Current e-commerce applications already provide product discovery, recommendations, ratings and reviews, review summaries, price comparison or price tracking, and conversational shopping assistance. Amazon's current Alexa for Shopping supports product research, comparisons, recommendations, review highlights and price-history features [4]. Google Shopping supports product discovery across sellers, reviews and ratings, price tracking and price insights [5]. Keepa focuses on Amazon price histories, product data and price-drop alerts [6]. These systems show that modern shopping applications increasingly assist users in evaluating products, but their documented features are generally organized around discovery, comparison, review information or price tracking rather than a single consumer-risk assessment pipeline.

ii. What Munasib Khareed Brings

- A virtual Consumer Advocate focused on evaluating a selected product rather than simply recommending products.
- A unified analysis of product metadata and customer-review evidence, including sentiment, product aspects, rating distribution, helpfulness, verified-purchase information and temporal review patterns.
- A Consumer Risk Assessment mechanism that combines product quality, evidence strength, rating reliability, negative-feedback concentration and price position.
- Fuzzy Logic to represent uncertainty when consumer evidence is incomplete, conflicting or unclear.
- TOPSIS-based multi-criteria decision analysis to combine the selected consumer criteria into a structured decision score.
- SHAP-based Explainable AI to identify and display the factors contributing to the consumer-risk assessment.
- Alternative Product Intelligence that evaluates related products using the same evidence-oriented decision-support approach.
- A Consumer Advocate Dashboard that presents product evidence, risk score, evidence strength, major concerns, explanations, alternatives and the final Consumer Verdict in one interface.
- A research-oriented evaluation of the AI/NLP components and the integrated decision-support system using measurable performance, consistency and usefulness checks.

iii. Research Gap Addressed

The project addresses the gap between product discovery/recommendation tools and evidence-based purchase-risk assessment. The contribution is not to replace existing shopping platforms, but to organize

heterogeneous product and review evidence into an interpretable decision-support pipeline: Data → Intelligence → Evidence → Risk → Decision → Explanation → Alternatives → Consumer Verdict.

f. Comparative Analysis of Existing E-Commerce Applications

The following comparison summarizes documented capabilities of selected current applications and shows how the proposed Munasib Khareed system differs in its consumer-risk and evidence-analysis focus.
The comparison indicates the intended contribution of Munasib Khareed: instead of competing with existing platforms on product discovery alone, the project focuses on evaluating the strength and consistency of consumer evidence and converting that evidence into a structured consumer-risk assessment and explainable Consumer Verdict. This makes the system a complementary decision-support layer centered on the question of whether the available evidence supports purchasing a selected product.

3. Problem Statement

E-commerce consumers are presented with an amount of product information. This information is spread across product metadata, ratings, customer reviews, helpfulness votes, purchase verification, prices, rankings and related products. Because of this spread E-commerce consumers may have to look at and understand a big amount of different evidence before deciding on a purchase.
Current recommendation systems mainly focus on finding or ranking products. They do not systematically look at the strength, consistency and possible risks that come from the consumer evidence, about a chosen product.
The main goal of this research is to create an AI framework that can assess consumer risk and help E-commerce consumers make better purchase decisions by combining different kinds of product and review evidence.
The proposed system will study how to use NLP, machine learning, Fuzzy Logic, TOPSIS and Explainable AI together. The aim is to turn product data and review data into a consumer-risk assessment and useful purchase guidance.

4. Motivation

The motivation behind Munasib Khareed is to provide consumers with a more informative approach to evaluating products before purchasing them.
The project aims to:

- Reduce the manual effort required to analyze large numbers of product reviews.
- Extract meaningful information from customer feedback.
- Identify positive and negative product aspects.
- Analyze the reliability and strength of available consumer evidence.
- Identify potential consumer risks associated with a product.
- Handle uncertainty in product and review evidence.
- Combine multiple decision criteria into a structured assessment.
- Provide explanations for the generated consumer-risk score.
- Present major concerns in an understandable format.
- Identify relevant alternative products.
- Provide consumers with a simple dashboard for purchase decision support.

a. Aims and Objectives

i. Aim

The main goal of the project is to create an AI-powered, easy-to-understand e-commerce decision-support system. This system will look at product details and customer reviews. It will check for risks that consumers might face when buying a product. Then it will give simple advice on whether or not to make a purchase. The focus is, on making sure the decisions are explainable and trustworthy.

ii. Objectives

The objectives of this project are listed below

- To analyze product metadata from the chosen UCSD Amazon Reviews dataset.
- To perform sentiment analysis on customer reviews with NLP and machine-learning techniques.

- To analyze rating distributions and review helpfulness.
- To examine verified-purchase information and temporal review patterns.
- To extract key positive and negative product aspects from reviews.
- To develop a consumer-risk assessment mechanism based on multiple evidence indicators.
- To use Fuzzy Logic to handle uncertainty in consumer evidence.
- To use TOPSIS for multi-criteria consumer decision analysis.
- To apply SHAP to explain the factors contributing to consumer-risk assessment.
- To identify alternative products with related-product information.
- To develop a Consumer Advocate Dashboard.
- To generate an interpretable final Consumer Verdict, from the analyzed evidence.

b. Scope of the Project

The scope of Munasib Khareed: AI-Driven E-Commerce Decision Support System outlines the features, functions, activities and deliverables needed to build an AI consumer decision-support system. The project will use a product category from the existing UCSD Amazon Reviews dataset, not the entire Amazon dataset. The system will examine product details and consumer feedback to spot consumer risks judge how strong the evidence is explaining what factors influence each assessment and give clear help with purchase decisions. The project will consist of the following components:

i. Product Intelligence

Product Intelligence module is a tool that examines the product-level data that's present, in the dataset. It looks at the product title, metadata, brand, category, description, features, price, sales rank and related products. By studying these attributes, the Product Intelligence module builds a picture of the chosen product and supplies useful data for later risk and decision-analysis tasks.

ii. Review Intelligence

Review Intelligence module is a system that handles customer reviews by applying Natural Language Processing and machine-learning methods. When deep-learning NLP is suitable the Review Intelligence module uses PyTorch to build and train models. The Review Intelligence module examines the review text, sentiment, rating spread how helpful the review's whether the purchase was verified and when the review was posted. It also picks out positive and negative aspects of the product that customers mention.

iii. Customer Risk Assessment

The Consumer Risk Assessment section will look at signs that come from the product and the reviews. These signs will include how good the product is, how strong the reviews are, how reliable the ratings are, how negative feedback there is and other factors that are supported by evidence. This part will bring all these signs together to create a risk rating for the product that is chosen.

iv. Decision Orchestration

The Decision Orchestration part will bring together signs from the consumer data to help make the final choice. Fuzzy Logic will be used to deal with uncertainty and unclear information in the consumer data while TOPSIS will be used for making decisions based on different factors. The information and the assessments will be put together to help create a clear and easy to understand decision.

v. Explainable AI

The Explainable AI part will make it clear what factors are affecting the systems decision. SHAP will be used to show which features are important and to make it easy to see them. The system will also explain the risk factors show the evidence from the reviews and give reasons, for the decision so that people can understand why a certain consumer assessment was made.

vi. Alternative Product Intelligence

The Alternative Product Intelligence module will find alternatives by looking at the related-product connections in the dataset. This includes products that're similar bought together or viewed together when that information exists. These products will be. Shown as possible options to help with comparing and making smart choices.

vii. Consumer Advocate Dashboard

The Consumer Advocate Dashboard will offer a way for users to pick and look into products. The dashboard will show the product summary, review breakdown, consumer-risk score, evidence-strength check, main issues, related alternative products and the final Consumer Verdict in a way.

The overall range of the system will therefore include the process from choosing a product looking at evidence checking risks explaining things finding alternatives and helping with the final buying decision.

viii. Limitations

To keep the focus and make the project manageable for the Final Year Project the following features will not be part of the system:

- Real-time web scraping from online stores.
- Complete fake-review detection.
- AI-based price talks.
- Predicting product life cycles.
- Using computer vision or image analysis for products.
- Unrelated parts of a recommendation system.
- Creating a product database.
- LSTM-based price prediction, unless the chosen dataset has a good continuous history of prices that can be used for this analysis.

The project will mainly concentrate on looking at a product like a consumer advocate measuring the evidence, from customers checking the risks explaining why the assessment was made and giving info to help the customer make a decision.

5. Proposed Methodology

The proposed methodology for Munasib Khareed: AI-Driven E-Commerce Decision Support System will be discussed. This method looks at product information and consumer feedback in an organized way. It turns the evidence we gather into a consumer‑risk assessment that people can understand easily. It helps them decide what to buy. The approach mixes steps. Data preprocessing comes first followed by product intelligence. Then we use Natural Language Processing and deep learning, with PyTorch. After that we evaluate consumer risk use Fuzzy Logic and run a TOPSIS‑based multi‑criteria analysis. We add AI with SHAP to explain the results. We also compare products and show everything on a dashboard for easy viewing.

The proposed methodology consists of the following stages:

a. Dataset Selection

We will use the existing UCSD Amazon Reviews dataset. Of using the whole Amazon data I will pick one easy product group to keep the project manageable and focused.

The selected category will be studied to find the availability of product and review details including:

- Product title and metadata
- Brand and category
- Product description and features
- Price information
- Sales rank
- Customer reviews
- Ratings
- Helpful votes
- Review timestamps
- Verified-purchase information
- Related also-bought or also-viewed products

The selected category will be the main source of data, for the proposed system.

b. Data Preprocessing

We will run Data Preprocessing on the chosen dataset before it reaches the analysis modules. Data Preprocessing aims to make the data consistent and usable for machine learning NLP and decision making.

The preprocessing procedure will include:

- Load the selected Amazon dataset.
- Inspect the product and review attributes that're available.
- Handle any incomplete values as needed.
- Remove duplicate or unsuitable records when appropriate.
- Clean the customer review text.
- Convert the text into a format that NLP models can use.
- Prepare the attributes such as ratings, price, helpful votes and sales rank.
- Process the review timestamps for temporal analysis.
- Prepare the purchase information.
- Construct the final feature set that the next modules need.

After Data Preprocessing we will send the processed data to the Product Intelligence and Review Intelligence modules.

c. Product Intelligence Analysis

The Product Intelligence module will analyze product level information to build a profile of the chosen product. The module will analyze:

- Product metadata
- Brand
- Category
- Product description. Features
- Price
- Sales rank
- Related products

The extracted information will give contextual evidence about the product and will be used together with review based evidence, in the consumer risk assessment.

d. Review Intelligence Using NLP and PyTorch

The Review Intelligence module will look at customer reviews using Natural Language Processing and machine-learning methods.

PyTorch will be used when deep learning is needed for Natural Language Processing. A good neural NLP model will be picked based on the type of data in the dataset. Possible options are a PyTorch-based LSTM or GRU model or a suitable Transformer model that is built or adjusted using PyTorch. The steps are shown in "Fig 1"

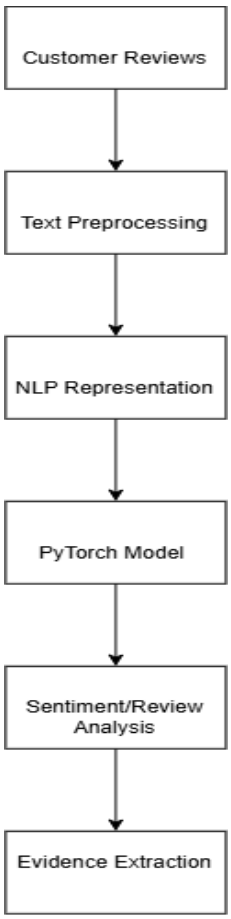


Fig 1. Flowchart for Review Intelligence using PyTorch and NLP

The module will do the following analysis:

- Process the text from customer reviews.
- Create representations of the text.
- Check the sentiment. Classify it.
- Find out what customers like and dislike.
- Pick out parts of the product where possible.
- Look at how sentiment connects with ratings.
- Check how helpful the reviews are.
- Look at verified-purchase data.
- Check if sentiment in reviews changes over time.
- Make evidence from reviews for the Consumer Risk Engine.

The performance of the chosen NLP model will be checked with measures like accuracy, precision, recall and F1-score.

e. Rating and Review Evidence Analysis

The system will use review-related signs instead of just using the average product rating.
The analysis will look at:

- How ratings are spread out
- How positive and negative sentiment is spread out
- How many reviews there are
- How helpful the reviews are
- If the purchase was verified
- How reviews change over time
- If negative feedback is all in one place
- What the positive and negative parts of the product are

These signs will be turned into useful features, for the Consumer Risk Assessment module.

f. Consumer Risk Assessment

The Consumer Risk Assessment module will look at the information gathered from product and review analysis.
The module will create signs related to:

- Product Quality
- How strong the review evidence is?
- How reliable the ratings are?
- How negative feedback there is?
- Overall consumer risk

The risk assessment will not just use one sign. Instead it will use signs based on evidence to make a better and more complete assessment of the product. The steps are shown in "Fig 2"

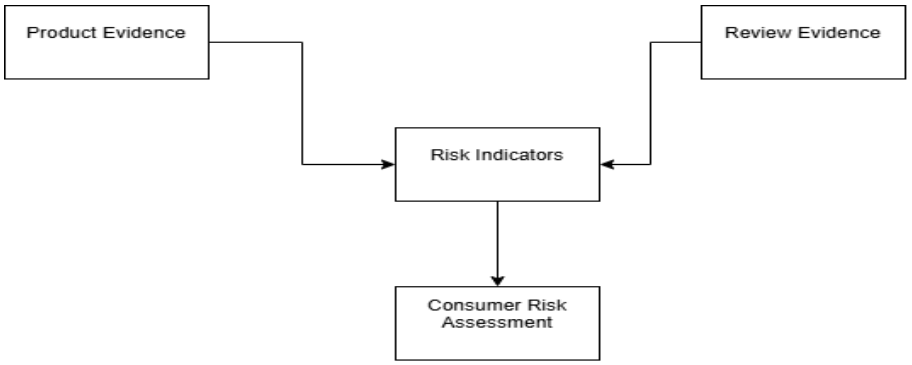


Fig 2. Flowchart for Customer Risk Assessment

The signs that come out will go to the Decision Orchestration part.

g. Fuzzy Logic-Based Uncertainty Handling

Consumer evidence does not always offer an answer. It can be uncertain. A product might have an average rating but also get a lot of negative comments. This makes it hard to decide if the product is good or bad. To deal with this fuzzy logic will be used in the decision process.
Fuzzy logic works with unclear or partial information. It helps make decisions even when the data's not certain. The model will take in inputs. These include product quality, evidence strength, rating reliability, negative-feedback concentration and price position. Each of these inputs will be turned into values using membership functions. These functions show how much each input belongs to a category. Like low, medium or high.
Then fuzzy rules will be applied. These rules connect the inputs to the idea of consumer risk. For example, if product quality is high but negative feedback is strong the system might say there is risk. The result is a risk score. This score is not a number. It is a range or degree of risk. This fuzzy assessment will later be used in the decision process.

h. TOPSIS-Based Decision Orchestration

The system will use TOPSIS to help make decisions based on criteria. TOPSIS stands for Technique for Order Preference by Similarity to Ideal Solution. It compares options by how close they're to the best possible choice and how far they are from the worst.

- The decision process starts with a matrix. This matrix lists all the products and the criteria for each. The criteria include product quality, review evidence strength, rating reliability, negative-feedback concentration and price position. These criteria are chosen based on the data set and results from experiments.
- Next the matrix is normalized. This means all values are scaled to a range. Then weights are applied to each criterion. These weights reflect how important each factor is. For example, product quality might be more important than price.

- After that the positive ideal solution and the ideal solution are found. The positive ideal is the possible value for each criterion. The negative ideal is the worst. Then the distance of each product from these points is calculated. The closer a product is to the ideal and farther from the negative ideal the better it is.
- Finally, the relative closeness to the solution is computed. This gives a decision score for each product. This score will help rank the products. It will go into the decision support system.

i. Consumer Risk Aggregation

All the information from the parts. Review Intelligence, Product Intelligence, Fuzzy Logic and TOPSIS. Will be put together in the Consumer Risk Engine. This engine will combine the outputs into one consumer-risk score.
The score will be on a scale, such as 0 to 100. For example, a score of 67/100 means there is risk. This score will not be shown alone. It will come with supporting details. These details will include scores for product quality review evidence strength, rating reliability and price position.
The system will also list any risk factors. These could be things like a lot of feedback or low reliability in ratings. This way users get both a number and the reasons, behind it.
This approach gives a picture. It shows not the final risk level but also the evidence that led to it. This helps consumers make choices.

j. Explainable AI using SHAP

Explainability is a key part of the system. It will use SHAP (SHapley Additive exPlanations) to show how each feature affects the model's output.
The Explainable AI process will include:

- Choosing the model or decision part to explain.
- Giving the feature values needed for the explanation.
- Figuring out how much each feature contributes.
- Spotting what makes the consumer-risk score go up or down.
- Showing the important factors in a clear way.
- Linking those factors to product details or review evidence.

For example, the system might say that a high risk score came from a lot of reviews worse sentiment lately or weak evidence from the data.
The goal is to not give a final answer but to also explain why that answer was reached. This helps users trust the system more.

k. Alternative Product Intelligence

After checking the chosen product, the system will find alternatives using connections, in the dataset.
It may look at:

- Similar products
- bought products
- Also-viewed products
- Related products

Each alternative will be checked using the same product and review data. The idea is not to turn this into a recommendation engine but to offer extra choices when the original product has serious issues or when comparing options helps the user make a better decision.

l. Consumer Verdict Generation

The last step is to combine everything from parts and create a clear Consumer Verdict.
Possible verdicts could be:

- CONSIDER
- CONSIDER WITH CAUTION
- CONSUMER RISK

The exact points where one verdict turns into another will be set during development and testing.
Major Concerns:

- Recent reviews show negative feelings.
- Customer feedback is not consistent.

- The evidence is not strong enough to say the product is safe to buy without worry.

All these numbers and notes are pulled from the data and rules built into the system not added by hand.

m. Dashboard and Result Visualization

The final results will be shown through a Consumer Advocate Dashboard.
The dashboard will provide:

- Product overview.
- Product metadata.
- Rating distribution.
- Review sentiment analysis.
- Positive and negative aspects.
- Review evidence strength.
- Consumer-risk score.
- Concerns.
- SHAP-based explanations.
- Relevant alternative products.
- -Final Consumer Verdict.

The dashboard will let users grasp the evidence without having to look at individual reviews one by one.

n. Overall Proposed System Procedure

The complete proposed methodology can be summarized through the following procedure:

1. Select a product category from the UCSD Amazon Reviews dataset.
2. Preprocess product and review data.
3. Allow the user to select a product, from the dataset.
4. Retrieve the selected products metadata, reviews, ratings, price, sales rank and related-product information.
5. Analyze product information using the Product Intelligence module.
6. Process customer reviews using NLP and PyTorch-based models where
7. Generate sentiment, aspect, rating, helpfulness, verification and temporal review features.
8. Calculate consumer-evidence indicators.
9. Apply Fuzzy Logic to handle uncertainty within the evidence.
10. Apply TOPSIS to perform -criteria decision analysis.
11. Aggregate the resulting indicators to calculate the Consumer Risk Score.
12. Apply SHAP to explain the factors contributing to the assessment.
13. Identify alternative products using related-product relationships.
14. Generate the Consumer Verdict.
15. Display the product analysis, evidence, risk assessment, explanations, concerns and alternatives through the Consumer Advocate Dashboard.

o. Proposed System Workflow

The overall System Workflow can be seen in "Fig 3"

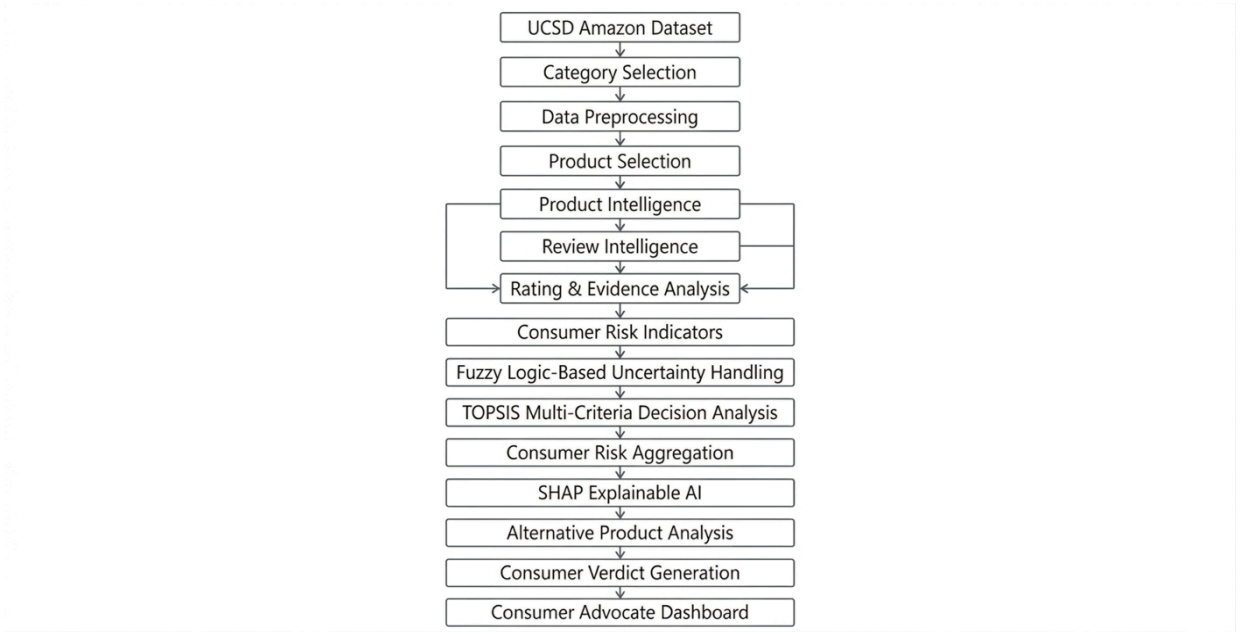


Fig 3. Workflow of the Overall System

p. Proposed Algorithm

Algorithm 1: AI Consumer Advocate Decision Process
Input: Selected product (P) product metadata, reviews, ratings, price, sales rank and related products.
Output: Consumer Risk Score, Evidence Strength, Major Concerns, Alternatives and Consumer Verdict.
Procedure:

1. Load the selected product (P) from the processed dataset.
2. Retrieve the product metadata and customer reviews.
3. Preprocess the retrieved product and review information.
4. Extract product-level features.
5. Process review text using NLP techniques.
6. Apply the selected PyTorch-based NLP model for sentiment/review analysis.
7. Calculate review and evidence indicators.
8. Calculate product quality and other relevant product indicators.
9. Construct the consumer-risk criteria.
10. Apply Fuzzy Logic to model uncertainty in the evidence.
11. Construct the TOPSIS decision matrix.
12. Normalize the decision matrix and apply the selected criteria weights.
13. Calculate the negative ideal solutions.
14. Calculate the TOPSIS decision score.
15. Aggregate the decision indicators into the Consumer Risk Score.
16. Apply SHAP to determine the contribution of features.
17. Retrieve and analyze relevant alternative products.
18. Generate the Consumer Verdict based on the defined decision rules.
19. Display the results and explanations through the Consumer Advocate Dashboard.
20. End.

q. Methodology Integration

The proposed methodology integrates computational techniques according to their specific roles. PyTorch will be primarily used for deep-learning-based NLP tasks, customer-review analysis. Fuzzy Logic will address uncertainty in consumer evidence while TOPSIS will combine decision criteria. SHAP will provide explanations of model outputs and important risk factors.

The resulting system will therefore follow the methodology:
Data → Intelligence → Evidence → Risk → Decision → Explanation → Alternatives → Consumer Verdict
This methodology is intended to transform heterogeneous e-commerce information into an interpretable decision-support output while keeping the project focused on consumer risk assessment rather, than conventional product recommendation.
The proposed Decision Making Process is shown in "Fig 4"

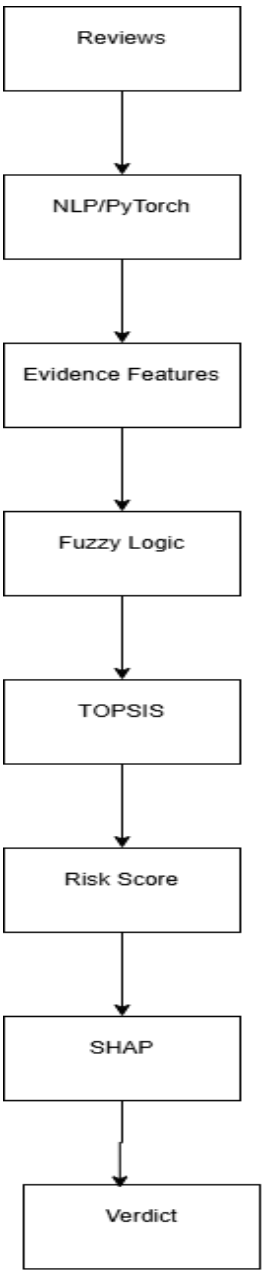


Fig 4. Proposed Decision Making Process

6. System Architecture

The System Architecture Diagram is shown in "Fig 5" below.

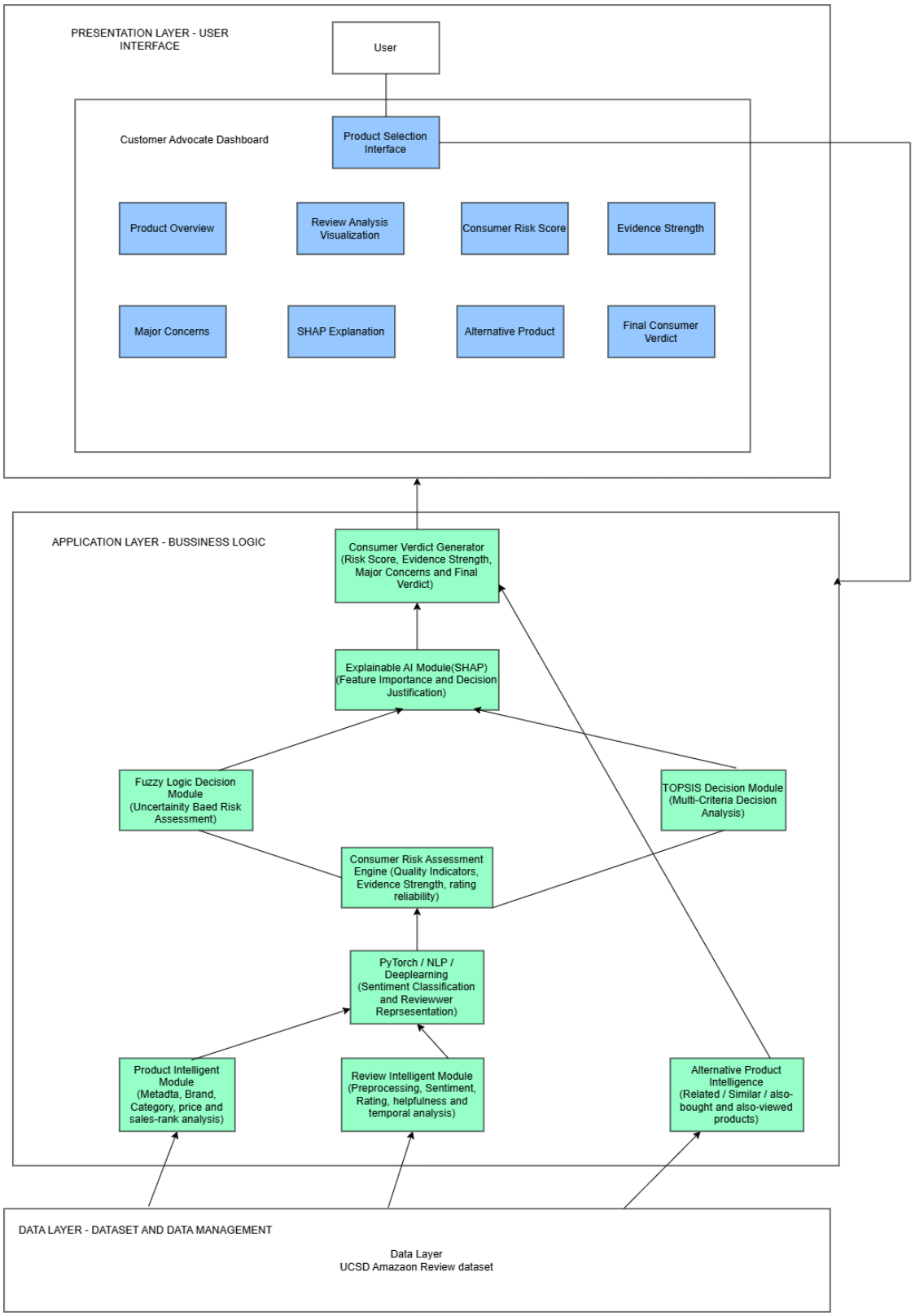


Fig 5. System Architecture Diagram

7. Individual Tasks

The individual tasks breakdown is present in "TABLE 2"

TABLE 2. WORK BREAKDOWN SCHEDULE

Team Member	Activity	Tentative Date
Abuzar Ali	Literature review and requirement analysis	28 Sep – 10 Oct 2026
	Dataset selection and exploration	11 Oct – 20 Oct 2026
	Dataset preprocessing	21 Oct – 10 Nov 2026
	Product Intelligence Module	11 Nov – 10 Dec 2026
	System integration	1 Mar – 31 Mar 2027
	Final testing and documentation	1 Apr – 26 Apr 2027
Muhammad Shehryar Choudhary	Literature review and NLP research	28 Sep – 15 Oct 2026
	Review preprocessing pipeline	16 Oct – 10 Nov 2026
	Sentiment analysis using PyTorch	11 Nov – 20 Dec 2026
	Aspect extraction and review intelligence	21 Dec 2026 – 31 Jan 2027
	Model evaluation and optimization	1 Feb – 20 Feb 2027
Muhammad Maaz Nizami	Literature review and decision-method research	28 Sep – 15 Oct 2026
	Consumer Risk Engine	16 Nov – 20 Dec 2026
	Fuzzy Logic implementation	21 Dec 2026 – 15 Jan 2027
	TOPSIS decision module	16 Jan – 10 Feb 2027
	SHAP and Explainable AI	11 Feb – 28 Feb 2027
	Dashboard and visualization	1 Mar – 31 Mar 2027

8. Gantt Chart

The Gantt Chart for Work Breakdown Schedule is shown in "Fig 6"

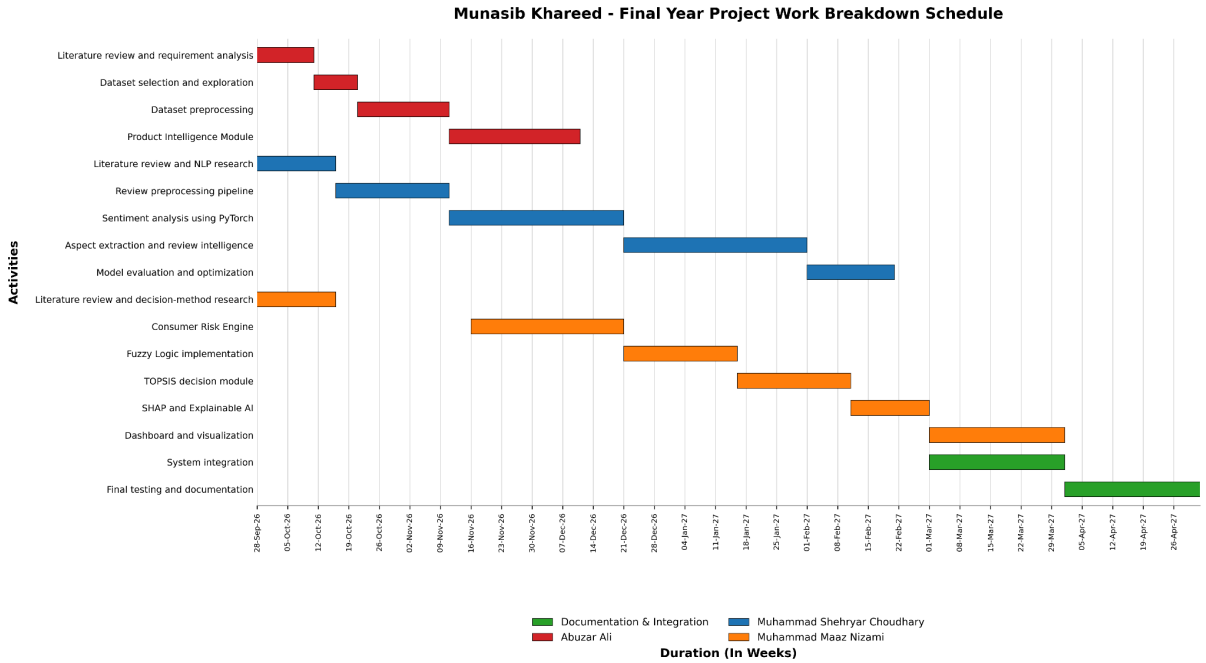


Fig 6. Gantt Chart for Work Breakdown Schedule

9. Tools and Technologies

The tools and technologies which we will use in this project are present in "TABLE 3"

TABLE 3. TOOLS AND TECHNOLOGIES

Category	Technology
Programming Language	Python
Deep Learning	PyTorch
NLP	PyTorch, Transformers / NLTK / spaCy
Machine Learning	Scikit-learn
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Plotly
Fuzzy Logic	scikit-fuzzy
Decision Making	TOPSIS
Explainable AI	SHAP
Dashboard	Streamlit
Development Environment	Jupyter Notebook, VS Code
Dataset	UCSD Amazon Reviews Dataset
Version Control	Git / GitHub

10. Expected Deliverables

This project aims to create a working and understandable AI system that helps consumers evaluate products sold online. The main deliverables are listed below.

a. Preprocessed Dataset

Preprocessed Dataset is a cleaned and structured subset of the UCSD Amazon Reviews data for the chosen product category. Preprocessed Dataset will contain product. Review details needed by the system, such as ratings, review text, helpfulness, dates, verification, price, sales rank and related products when available.

b. Product Analysis Component

Product Analysis Component can. Examine the basic details of the chosen product, such as title, category, brand, description or features, price, sales rank and any related products.

c. Review Analysis Component

Review Analysis Component reads customer reviews. Produces useful insights for example sentiment, rating distribution, key positive and negative points, helpfulness patterns, verification status and trends over time.

d. Consumer Risk Assessment Component

Consumer Risk Assessment Component turns product and review data into risk signals for consumers. It gives a risk score plus signals such as product quality, evidence strength, rating reliability and concentration of negative feedback.

e. Decision Support Component

Decision Support Component combines consumer criteria and uncertainty data to create a clear assessment of the chosen product.

f. Explainability Component

Explainability Component shows the factors that influence the consumer risk assessment. It offers explanations instead of just a number.

g. Alternative Product Analysis

Alternative Product Analysis. Shows relevant alternative products from the related-product data. These alternatives help consumers compare the chosen product with options.

h. Consumer Advocate Dashboard

Consumer Advocate Dashboard brings together the outputs of the system. It displays product details review insights, risk assessment, evidence strength, key concerns, explanations, alternatives and the final consumer verdict.

i. Consumer Verdict

Consumer Verdict is the consumer-friendly decision that comes from the integrated analysis. Depending on the evidence the system may say Consider, with Caution or Avoid, along with the reasons.

j. System Evaluation Results

System Evaluation Results show data that tests the AI components and the whole decision-support system. It includes performance measures, model comparisons and checks of consistency and usefulness.

k. Complete Integrated System

Complete Integrated System is a working prototype that combines the dataset, product analysis, review analysis, risk assessment, decision support, explainability, alternative analysis and dashboard into one AI Consumer Advocate system.

l. Project Documentation

Project Documentation contains all information, literature review, dataset details, methods, system design, implementation, experiments, testing, evaluation, results, limitations and ideas, for future work.

m. Final Demonstration and Presentation

Final Demonstration and Presentation shows the finished system using selected products from the dataset and includes the project talk and supporting materials.

i. FYP-I Deliverables (Deadline: December 2026)

- Literature review and requirements analysis for AI-driven e-commerce decision support and consumer evidence analysis.
- Selection and exploration of one suitable product category from the existing UCSD Amazon Reviews dataset.
- Data preprocessing pipeline covering product metadata, reviews, ratings, helpful votes, timestamps, verified-purchase information, price, sales rank and related products where available.
- Initial Product Intelligence module for extracting and organizing product-level information.
- Review preprocessing and NLP pipeline for customer-review analysis.
- Initial sentiment analysis using a suitable PyTorch-based NLP model, with baseline evaluation using accuracy, precision, recall and F1-score.
- Initial aspect extraction and Review Intelligence features for identifying positive and negative product aspects.
- Initial Consumer Risk Assessment mechanism and risk indicators based on product and review evidence.
- FYP-I technical documentation, preliminary experiments and progress evaluation leading into FYP-II.

ii. FYP-II Deliverables (Deadline: May 2027)

- Fuzzy Logic-based uncertainty handling and consumer-risk aggregation.
- TOPSIS-based multi-criteria decision module and decision-score generation.
- SHAP-based Explainable AI and interpretation of consumer-risk factors.
- Alternative Product Intelligence and evidence-based comparison of related products.
- Consumer Advocate Dashboard integrating product information, review insights, risk assessment, evidence strength, concerns, explanations and alternatives.
- Integrated Consumer Verdict generation based on the defined decision rules.
- System integration across Product Intelligence, Review Intelligence, Consumer Risk Assessment, Decision Orchestration, Explainable AI, Alternative Product Intelligence and Dashboard modules.
- Model evaluation, optimization and complete system testing using defined performance and consistency measures.
- Final documentation covering implementation, experiments, testing, evaluation, results, limitations and future work.
- Final demonstration and presentation of the complete integrated Munasib Khareed prototype.

11. References

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